

Weekend Healthcare Pulse: Japan's drug pricing scheme irks pharma companies; may eventually hurt patients' interests

Miki Sogi, Ph.D. +81 3 6777 6991 miki.sogi@bernsteinsg.com

Rebecca Liang, Ph.D. +852 2123 2656 rebecca.liang@bernsteinsg.com

Justin Smith +44 20 7762 5899 justin.smith@bernsteinsg.com

Eve Burstein +1 917 344 8313 eve.burstein@bernsteinsg.com

Lee Hambright +1 917 344 8429 lee.hambright@bernsteinsg.com

Susannah Ludwig +41 582 723 127 susannah.ludwig@bernsteinsg.com

Lance Wilkes +1 917 344 8501 lance.wilkes@bernsteinsg.com

Delphine Le Louet +33 1 42 13 92 93 delphine.le-louet@bernsteinsg.com

Courtney Breen +1 917 344 8407 courtney.breen@bernsteinsg.com

William Pickering, MD +1 917 344 8340 william.pickering@bernsteinsg.com

Nandan Kulkarni +91 22 6842 1436 nandan.kulkarni@bernsteinsg.com

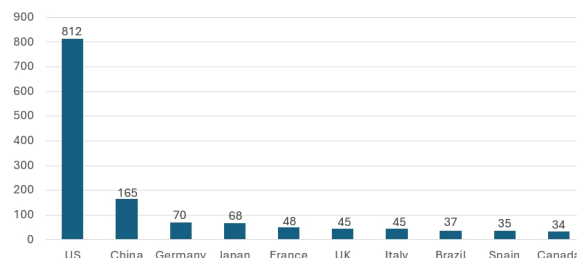
Jeffrey Walch +1 917 344 8613 jeffrey.walch@bernsteinsg.com

LOW DRUG PRICES IS MAKING JAPAN PHARMACEUTICAL MARKET UNATTRACTIVE AND JAPANESE COMPANIES' OVERSEAS BUSINESS EXPANSION IMPERATIVE FOR GROWTH

Today Japan is the fourth pharmaceutical market based on drug sales and less than a tenth size of the US market, tumbling from being second in global ranking with a market size a third of the US back in 2012 (Exhibit 1). Thus it is no surprise that Japan's pharmaceutical market growth has been the weakest among the key markets (Exhibit 2). One of the key drivers of Japan market's diminishing presence is low drug price—Japan's drug price is the lowest among the key markets when compared as % of the US price (Exhibit 3). Given this unattractive market situation, key Japanese pharma companies are expanding their overseas business to seek growth (Exhibit 4).

EXHIBIT 1: Today Japan is the fourth pharmaceutical market and less than a tenth size of the US, tumbling from being the second-biggest market with a third of the US's back in 2012

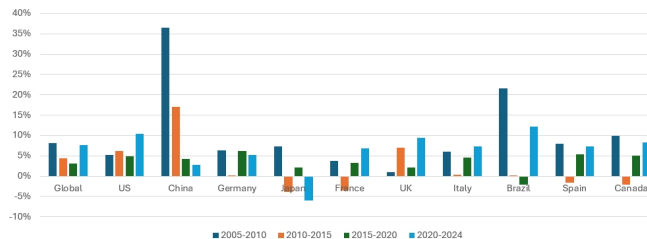
2024 global pharmaceutical market size in USD Bn



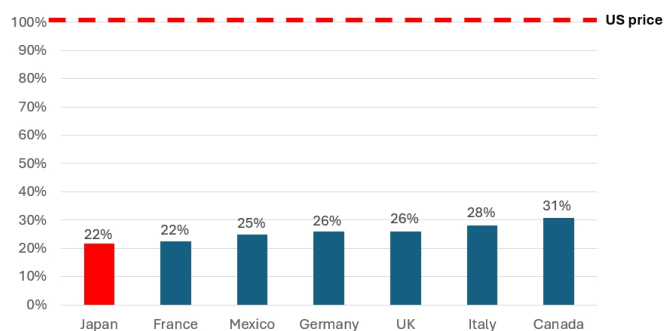
Source: Japan Pharmaceutical Industry Association, Bernstein analysis

EXHIBIT 2: Despite its aging population, Japan's pharmaceutical market is weakest among the key markets

Pharmaceutical market growth by country (CAGR %)

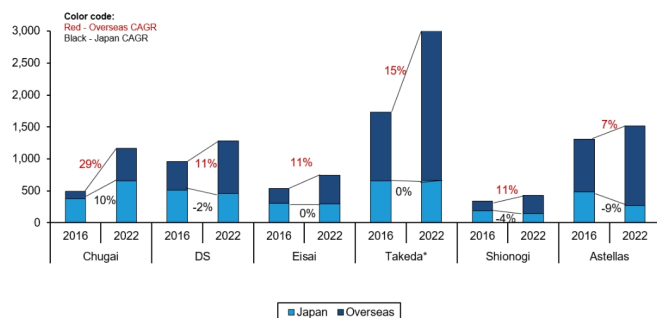


Source: Japan Pharmaceutical Industry Association, Bernstein analysis

EXHIBIT 3: Japan's drug pricing is the lowest among the peers**Branded drug listing price comparison as % of the US price**

Data from 2022

Source: RAND Health Quarterly, Bernstein analysis

EXHIBIT 4: Given the unattractive domestic market situation with low drug prices, key Japanese pharma companies are expanding their overseas business to seek growth**Revenue (JPY B) Breakdown by Japan vs. Overseas**

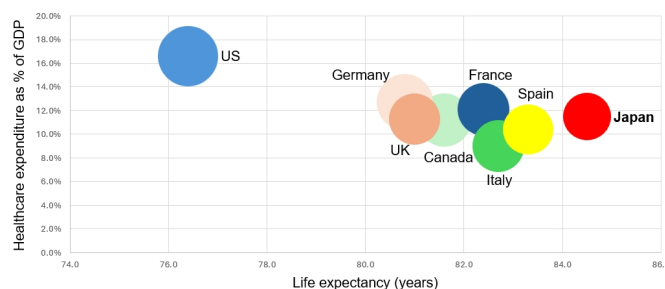
The majority of Takeda's growth comes from its acquisition of Shire in 2019; we don't have revenue forecast by region for Takeda & Shionogi

Source: Company reports, Bernstein analysis

IN JAPAN, DUE TO THE PRICING SCHEME, DRUG PRICE IS AN EASY LEVER FOR HEALTHCARE COST CONTAINMENT**JAPANESE GOVERNMENT HAS BEEN UNDER PRESSURE TO CONTAIN HEALTHCARE EXPENDITURE IN THE FACE OF FASTEST AGING POPULATION**

When measured based on the input and output, the productivity of Japan's healthcare system is arguably better than peer developed countries as Japan delivers the longest life expectancy without spending more in terms of % of GDP (Exhibit 5; we acknowledge that Japanese people's healthier diet and other factors - e.g., genetics, lifestyle - also contribute to their longevity). Yet Japan's healthcare system's productivity has been under enormous pressure due to its fast aging population, which is driving healthcare expenditure increase (Exhibit 6, Exhibit 7). Plus 38% of the cost is funded by the federal and local governments (Exhibit

8). So containing the healthcare spending has been one of the key priorities of the debt-ridden Japanese government.

EXHIBIT 5: The productivity of Japan's healthcare system is better than that of peer countries, delivering the longest life expectancy without spending more...

Life expectancy reported in 2021-2022

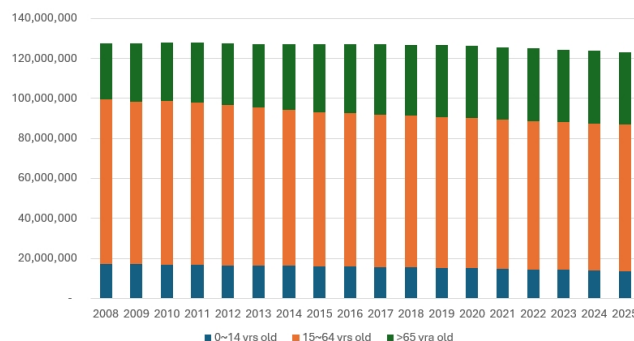
Health expenditure as % of GDP in 2022

The bubble size represents PPP-adjusted GDP per person

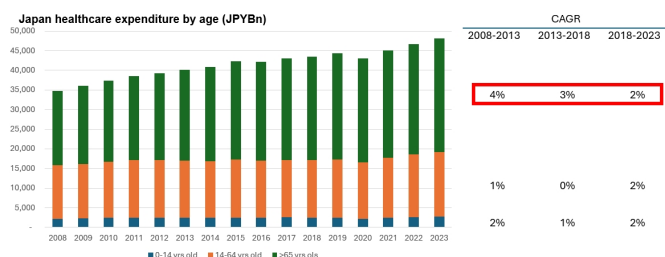
Source: OECD, World Bank, Our World in Data, Bernstein analysis

EXHIBIT 6: ... yet its fast-growing aging population is challenging healthcare expenditure...

Japan is the fastest aging society—% of >65 yrs old increased from 22% to 29% over the past 15 years



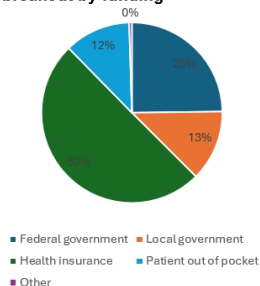
Source: Dashboard.E-stat.go.jp, Bernstein analysis

EXHIBIT 7: .. as the older population accounts for the majority of healthcare expenditure

Source: Japan Ministry of Health, Labor and Welfare, Bernstein analysis

EXHIBIT 8: In Japan, 38% of healthcare expenditure is funded by the federal or local government

Japan healthcare expenditure breakout by funding



Data in 2024

Health insurance funding represents employee contribution and the insured's premium

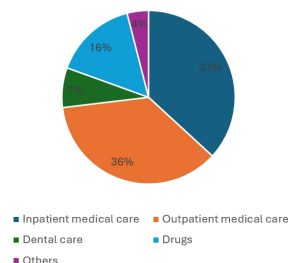
Source: Japan Ministry of Health, Labor and Welfare, Bernstein analysis

DRUG SPENDING IS AN EASY LEVER FOR THE JAPANESE GOVERNMENT TO PULL IN ORDER TO CONTAIN HEALTHCARE EXPENDITURE

While drug cost accounts for only 16% within Japan's overall healthcare expenditure, it was previously the fastest growth category until the mid-2010's (Exhibit 9, Exhibit 10). Thus, it became a clear target for the government's cost containment efforts. In Japan, the government has a full control of drug prices because almost all the drugs approved in Japan are reimbursed through the national health insurance (NHI, Japan has a universal health insurance system) and the government determines drugs' NHI price, i.e., the amount drug dispensers (i.e., pharmacies, hospitals, clinics) are reimbursed. There are a few exceptions, including erectile dysfunction drugs such as Viagra. On the other hand, reducing the cost of inpatient and outpatient medical care cost is politically difficult despite these categories accounting for 70%+ of the overall healthcare expenditure. For example, the Japan Medical Association, a clinic-based physicians group, is a powerful political constituency:

EXHIBIT 9: Drug spending only accounts for 16% of overall healthcare expenditure...

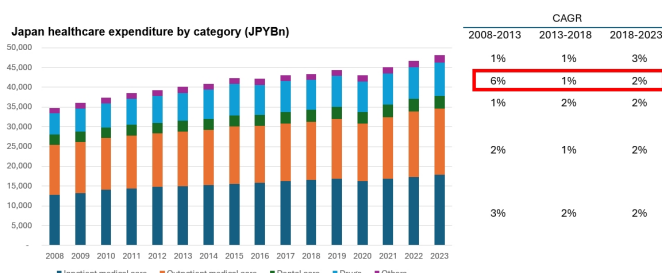
Japan healthcare expenditure breakout by category



Data in 2024

Source: Japan Ministry of Health, Labor and Welfare, Bernstein analysis

EXHIBIT 10: ... yet was the fastest growing category within the healthcare expenditure



Source: Japan Ministry of Health, Labor and Welfare, Bernstein analysis

IN JAPAN, THE GOVERNMENT HAS FULL CONTROL OF DRUG PRICING: PRICING AT LAUNCH AND DOWNWARD ADJUSTMENT WHILE IN THE MARKET

MOST NEW DRUG PRICING IS BASED ON COMPARISON WITH ANALOGOUS EXISTING DRUGS IN COMBINATION WITH A PREMIUM BASED ON INNOVATION, USEFULNESS, MARKET SIZE, ETC.

In Japan, Ministry of Health, Labor and Welfare (MHLW) grants new drug an official NHI price within 60 days after its regulatory approval. This price is the amount to be reimbursed to the drug dispensers (i.e., pharmacies, hospitals, clinics) under the national health insurance scheme. Pharma companies can start promoting the new drugs upon regulatory approval but have to wait for the NHI price to ship the products for distribution.

There are two approaches to new drug pricing: **comparison with analogous existing drugs or COGS-based** if a similar does not exist in the market.

In the **comparison approach**, a new drug's price is determined based on its "innovation" and "usefulness" vs the existing analogous

drugs. A drug would receive a **70–120% innovation premium vs the comparator** if meeting all the following three categories: clinically meaningful, novel MOA; higher efficacy and/or safety; and objectively proven improvement of the treatment approach for the target indications. A drug would receive a **5–60% usefulness premium** if it met one to two out of the above three categories or is objectively a more clinically useful formulation vs the comparator.

While 65% of the 639 drugs approved between 2018 and 2025 were priced based on the comparison approach, **the COGS-based approach** was applied to 26% of cases. This is trickier as pharma companies usually do not want to provide full disclosure of drug COGS. So this approach is more of an art than a solid data-based approach: MHLW has reported 60%+ drugs disclosed less than 50% of the COGS information for the pricing process for the recent years.

On top of the base pricing, there are multiple premium categories:

- Market premium, 5–20%: the drug targets a rare disease and is deemed an orphan drug.
- Pediatric premium, 5–20%: the drug has indications or dosing for pediatric use
- Frontier premium, 10–20%: the drug receives regulatory approval in Japan ahead of other countries

Japan looks to harmonize the domestic drug pricing with exJapan's to avoid criticism for its lower price vs peers. To eschew drugs' price excessive gap vs overseas standard, Japanese government uses average price of the US, UK, Germany and France. Despite this scheme, Japan's drug pricing is still the lowest, as low as that of France, among its peer developed countries and much lower than that in the US (Exhibit 3).

WHILE MANY DRUGS ARE SUBJECT TO ANNUAL PRICE CUTS, INNOVATIVE DRUGS ARE EXEMPTED

In Japan, drugs' NHI price is reviewed and adjusted down annually based on a principle that drug dispensers should not make more than 2% of profit margin from the gap between the drug purchase price from wholesalers and the NHI reimbursement amount. This concept is not unique to Japan: in France and Germany, pharmacies and wholesalers' margins are fixed by the government. Due to the competitive dynamics among wholesalers whose margin is neither protected nor limited in Japan, the dispensers' drug purchase price is usually under downward pressure, and the gap vs the NHI price tends to grow. Thus, the Japanese government conducts a market price survey every September and, based on the outcome, determines the adjustment amount to be disclosed the following March and implemented in April. For FY2026's adjustment, the survey revealed a 4.8% gap, which was the lowest in the past 30 years as it has been consistently declining, and the government has decided on a 0.9% price cut on average (the actual adjustment price is different by

drug). The price revision used to be every other year yet, in 2021, the government introduced an off-year revision, which has become part of the regular timeline.

Innovative drugs are exempted from the annual price cut until Gx launch or 15 years after the launch under the Patent-period Price Maintenance Program (PMP). The "innovativeness" of drugs is assessed based on criteria such as orphan drug status, recipient of innovation/usefulness premium, launch in Japan ahead of other markets, etc. The program was implemented to not discourage pharmaceutical companies from launching new innovative drugs in Japan. Under this program, most key growth drivers of global and top Japanese pharma companies should not be subject to the annual repricing. Of note, when the protection period expires, the drug's price gets cut at once to reflect all the repricing that could have happened without the PMP. Having said that, in most cases, Gx are in the market by that time, thus the lower price does not matter any longer (nowadays Gx penetration in Japan is as fast as in other developed countries).

OUTSIDE THE ANNUAL REPRICING, MHLW HAS AD-HOC TOOLS TO ADJUST DRUGS' NHI PRICE AFTER LAUNCH ON THE GROUNDS OF SALES AND COST-EFFECTIVENESS

Drugs with too much commercial success get punished in Japan. When a pharma company applies for an NHI price for a drug after the regulatory approval, pharma companies include in the application the drug's peak domestic sales forecast based on the approved indication. Under the Market Expansion Reassessment/ Sustainability Special Price Adjustment schemes, MHLW cut drugs' NHI price if their annual sales significantly exceed the initial peak domestic sales forecast: if a drug's annual sales reach >JPY100Bn and exceed 1.3~1.5x of the initial peak sales forecast, its NHI price can be cut by 10~50%. Within our coverage, Daiichi Sankyo's Lixiana, an anti-coagulant, has been subject to the cut twice, in 2018 by 28% and 2026 by 20%. This price cut is often inevitable, because the initial peak sales forecast included in the NHI price application only reflects the first approved indication. Thus the sales of drugs in therapeutic areas such as oncology and immunology, where most drugs are developed for multiple indications, can easily exceed the initial forecast over time by a wide margin. Previously, this price cut was applied not only to those drugs where sales exceed their forecast but also to those with the same mode of action (e.g., Keytruda and Opdivo). This rule was absolutely loathed by pharmaceutical companies, as you can imagine, and MHLW got rid of this rule in the most recent rule updates.

Cost-effectiveness based price adjustment has been fully incorporated in the official NHI pricing rules since 2020. Japan has applied cost-effectiveness-based price adjustments only to a relatively small, select set of high-cost drugs so far, mostly oncology and immunology products. The criteria of the selection are high innovation and large budget impact: typically drugs that had received significant innovation/usefulness premiums or

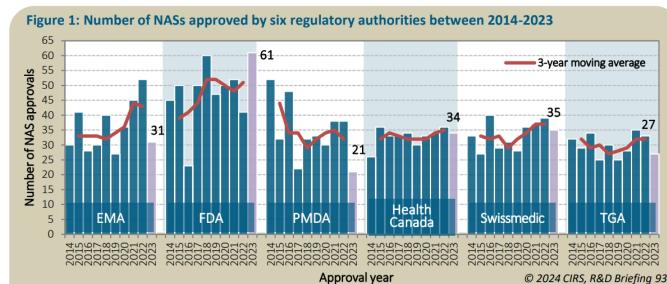
that generated very high peak sales. Within our coverage, Eisai's Leqembi for Alzheimer's disease received a 15% price cut in November 2025 on the grounds of cost-effectiveness. The drug was designated for cost-effectiveness evaluation soon after its NHI price listing because of its high unit price and expected large patient population in early Alzheimer's disease. An analysis of the incremental cost-effectiveness ratio (ICER) per quality-adjusted life year (QALY) showed a 25% price adjustment from the original, yet the maximum price reduction of 15% was applied.

LOW DRUG PRICES INDEED HELPS HEALTHCARE COST CONTAINMENT, BUT IS IT GOOD FOR PATIENTS?

Many US and EU pharma companies have been vocal about their discontent with the Japanese drug pricing system. While they don't like the lower launch price, the annual and ad-hoc price revisions are particularly incomprehensible and irk them. They have been warning the Japanese government that their unattractive drug pricing scheme will discourage them to develop and commercialize innovative drugs in Japan. We have yet to hear an outcry over not getting life-changing drugs in Japan that are available elsewhere; however, the number of new drug approvals in Japan has been in decline for the past decade while increasing in the US and the EU (Exhibit 11). R&D spending in Japan is also declining while increasing in the US, EU and China (Exhibit 12). While this can be partially explained by the fact that country-specific clinical trials are no longer required for regulatory approval in Japan (i.e., Japanese patient are included in global trials), it suggests that pharma companies' are decreasing investment in their drug development in Japan. Plus, the US's most favored nation drug price requirement for new products may get some pharma companies to opt out of launching in Japan in the future to protect revenue and profit from more lucrative markets, especially in the US. If this happens, the Japanese government will no longer be able to use drug prices as a

lever to manage healthcare spending.

EXHIBIT 11: # of new drug approvals has been declining in Japan while growing in the US and EU



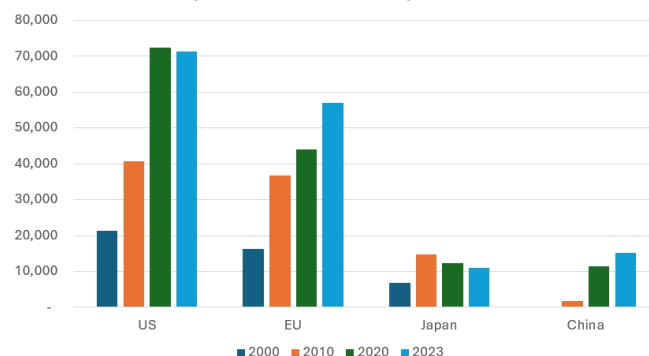
NAS: New Active Substance referring to new drug ingredient in EU.

EU EMA; US FDA; Japan PMDA; Australia TGA

Source: Centre for Innovation in Regulatory Science

EXHIBIT 12: R&D spend in Japan is declining while increasing elsewhere

Pharmaceutical R&D expenditure in the US, EU, Japan and China in USD Mn



Source: EFPIA, PhRMA, JPMA, China Statistical Yearbook, Bernstein analysis

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	28 May 2026	Price Target	TTM	Reported EPS				Reported P/E (x)		
			Closing Price		Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
4503.JP (Astellas)	M	JPY	2,124.00	2,200.00	8.9%	JPY	162.77	163.46	168.81	13.0	13.0	12.6
4519.JP (Chugai)	O	JPY	7,743.00	9,800.00	(37.5)%	JPY	263.73	316.80	373.10	29.4	24.4	20.8
4568.JP (Daiichi Sankyo)	O	JPY	2,662.50	4,500.00	(74.9)%	JPY	140.37	150.11	167.44	19.0	17.7	15.9
4523.JP (Eisai)	O	JPY	3,901.00	6,100.00	(43.4)%	JPY	163.76	168.31	211.50	23.8	23.2	18.4
4578.JP (Otsuka)	M	JPY	11,140	11,600	21.4%	JPY	685.06	563.74	634.04	16.3	19.8	17.6
4507.JP (Shionogi)	M	JPY	2,996.50	3,000.00	(17.0)%	JPY	241.11	247.57	264.99	12.4	12.1	11.3
4502.JP (Takeda)	O	JPY	5,120.00	6,900.00	(21.5)%	JPY	496.16	506.05	530.52	10.3	10.1	9.7
JPL			2,551.66									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

4502.JP estimate is Adjusted EPS; 4502.JP valuation is Adjusted P/E (x); 4523.JP base year is 2024;

Source: Bloomberg, Bernstein estimates and analysis.

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